English Medium

EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

LECTURE 11

NUTRITION-Part-02

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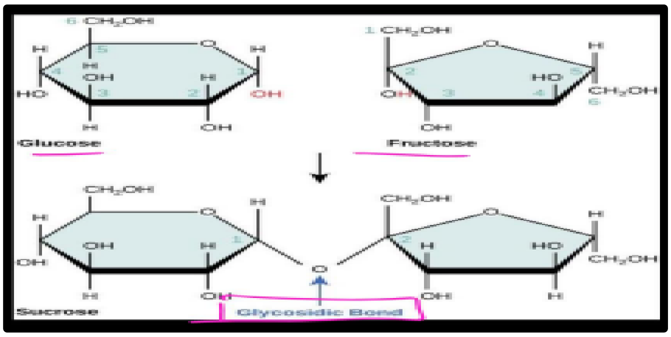
Major components of nutrient:

1. Carbohydrates
2. Protein
3. Fat

1. Carbohydrates

- Carbohydrates are organic compounds made up of carbon (C), hydrogen (H), and oxygen (O) atoms, typically in a 1:2:1 ratio, with the general formula Cx $(H_2O)_y$.
- They are one of the four major classes of biomolecules, alongside proteins, lipids, and nucleic acids.
- Carbohydrates are essential for life, serving as the primary source of energy in most organisms and playing structural and functional roles in cells.
- Main source of energy (55-65%)
- Glucose i.e., a type of carbohydrates acts as respiratory substrate.
  $6CO_2 + 6H_2O + \text{Light} \rightarrow C_6H_{12}O_6 + 6O_2$
- Carbohydrates are made up of Carbon, hydrogen and oxygen in the ratio of (1:2:1)
- Simple (carbohydrate which are sweet in taste known as sugar.
- On the basis of saccharide unit Carbohydrate are classified into—

I.Monosarcharide	II.Oligosacchoride	III.Polysaccharide
One molecule	2-10 molecule	100 thousand molecule
Eg.-Glucose Frutose, Galatose	**Eg.**-Maltose (Glucose+Glucose) **Sucrose:**(Glucose+frutose)	**Eg**-Starch Glycogen Cellulose



a) Monosaccharides

- Monosaccharides are the simplest form of carbohydrates, consisting of a single sugar molecule.
- They are the building blocks of more complex carbohydrates.
- Monosaccharides cannot be hydrolyzed into simpler carbohydrates.
- On the basis of number of carbon atom monosaccharides are of following types:
- **Trioses-** No. of Carbon atom=3 $(C_3H_6O_3)$
- **Teroses**–No. of Caron atom=4 no of corbon atom
- **Pentose—** no of carbon atom =5 $(C_5H_{10}O_5)$
- **Hexose—** No. of carbon atom=6 $(C_6H_{12}O_6)$
- **Heptose—** no of carbon atom=7 $(C_7H_{14}O_2)$
- Largest monosaccharide.
  E.g- Sudoheptulose
- The most common examples include:
  - **Glucose:** The primary source of energy in cells.
  - **Fructose:** Found in fruits and honey.
  - **Glucose,** being the most important monosaccharide, is the body's primary energy Follow Us of source
  - **Galactose:** A component of lactose found in milk.
  - **General formula:** $(CH_2O)_n$, where n is typically 3-7. Soluble in water and sweet in taste.

Disaccharides

- Disaccharides consist of two monosaccharide molecules linked by a glycosidic bond.
- They are broken down into their monosaccharide components during digestion.

Some common disaccharides are:

- Sucrose (glucose + fructose): Common table sugar.

- Lactose (glucose + galactose): The sugar found in milk.
- Maltose (glucose + glucose): Formed during the digestion of starch
- Soluble in water and sweet in taste.
- Must be broken down into monosaccharides by enzymes like lactase, sucrase, and maltase during digestion.

Polysaccharides

- Polysaccharides are complex carbohydrates made up of long chains of monosaccharide units.
- These are usually insoluble in water and do not have a sweet taste.
- Polysaccharides serve as energy storage molecules and structural components in plants and animals.

Examples include:

Starch:

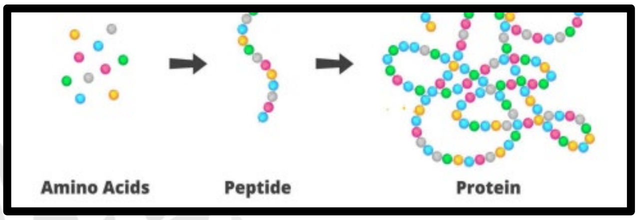
- The primary energy storage form in plants (e.g., potatoes, rice, wheat).
- Polymer of glucose → Made up of amylose 2 amylopectin. store food in plant.
- Gives online test.
- Amylose -Blue colour
- Amylopectin-1 Red colour.
  Eg- Potato
- **Glycogen:** The energy storage form in animals, stored mainly in the liver and muscles.
- **Cellulose:** A structural polysaccharide found in plant cell walls, providing rigidity and strength.
- **Chitin:** A component of the exoskeletons of insects and crustaceans.
- Polysaccharides are broken down into monosaccharides during digestion to release energy.
- Some polysaccharides, like cellulose, are indigestible by humans but serve as dietary fiber

2. Protein

Proteins are essential body-building nutrients required for growth, repair, and maintenance of body tissues. They are complex organic compounds made up of amino acids.

Amino Acids: -

- Amino acids are organic compounds containing an amino group [NH_2] and an acidic group [COOH] as substituents on the same carbon i.e., the α-carbon.
- Hence, they are called α-amino acids.
- They are substituted with methane.
- All proteins are polymers of α-amino acids



- Amino acids contain amino ($-NH_2$) and carboxyl (-COOH) functional groups.
- Depending upon the relative position of amino group with respect to carboxyl group, the amino acids can be classified as α, ß, y, 8 and so on.
- Only α-amino acids are obtained on hydrolysis of proteins.
- All α-amino acids have trivial names, which usually reflect the property of that compound or its source.
- Glycine is so named since it has sweet taste (in Greek glycol means sweet) and tyrosine was first obtained from cheese (in Greek, tyros means cheese.)
- The amino acids, which can be synthesized in the body, are known as nonessential amino acids.
- On the other hand, those which cannot be synthesized in the body and must be obtained through diet are known as essential amino acids.

3. Fat

- Fats are organic compounds and essential macronutrients composed of carbon, hydrogen, and oxygen, and they belong to the lipid family.
- Energy Source: Fats are the most concentrated source of energy, providing about 9 calories per gram, which is more than double the 4 calories per gram provided by carbohydrates or proteins.

- **Cell Structure and Hormones:** Lipids are essential components of cell membranes and are also involved in the production of certain hormones.
- **Essential Fatty Acids:** Some fatty acids are "essential" because the body cannot synthesize them and must obtain them through diet.

Types of Fats and Their Sources:

- Fats are categorized based on their chemical structure (presence or absence of double bonds) and their effects on health

Type of Fat	Health Implication	Common Sources
Saturated Fats	Considered unhealthy in excess, it can raise bad (LDL) cholesterol and increase heart disease risk.	Animal products like meat, butter, cheese, milk, and certain tropical oils (coconut, palm oil).
Trans Fats	The most harmful type of fat; artificially produced (partially hydrogenated oils) and strongly linked to heart disease.	Processed and packaged foods, fried foods, cookies, cakes, and some natural sources like meat and dairy in small amounts.
Monounsaturated Fats (MUFAs)	Healthy fats; help improve blood cholesterol levels and control inflammation.	Olive oil, avocado, peanuts, and nuts like almonds and hazelnuts.
Polyunsaturated Fats (PUFAS)	Healthy fats (including Omega-3 and Omega-6 fatty acids); essential for brain development, heart health, and immune function.	Vegetable oils, walnuts, flax seeds, and oily fishlike salmon and mackerel.

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